

ADEEL ASGHAR

Artificial Intelligence Undergraduate | Applied Machine Learning & Deep Learning

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SUMMARY

AI/ML undergraduate specializing in applied machine learning, deep learning, and AI engineering, with a strong academic and technical record validated by multiple hackathon and competition wins, including **1st place at HACKDATA V1**. GDG Tech Lead and founding member who has mentored **100+ students** in AI/ML fundamentals, with hands-on experience spanning computer vision, NLP/LLM systems, and full-stack AI deployment.

EXPERIENCE

AI Engineering Intern | CalderR

Jun 2026 – Present

- Resolved a BM25 keyword-bias failure in a personal-document RAG assistant by implementing hybrid retrieval with cross-encoder reranking, reaching 14/15 correctly grounded Q&A responses across 444 chunks from 31 source documents
- Built a real-time, multi-source research engine using a RAG pipeline with RAGAS-based relevance and faithfulness evaluation, generating automated reports beyond the 10-report delivery requirement while hardening output formatting for reliability
- Built four additional agentic AI tools — a CLI assistant, multi-agent research assistant, automated code-review agent, and data-analysis agent — covering LLM tool-calling and multi-step orchestration

AI Intern | Pakistan Ordnance Factories (POF)

Jul 2026 – Present

- Joined the AI team to support applied machine learning workflows, including data preprocessing and model evaluation tasks
- Ramping up on the team's production ML tooling and internal AI systems as part of an ongoing AI engineering rotation

Founding Member & Tech Lead | GDG On Campus - COMSATS Wah

Oct 2025 – Present

- Grew practical AI/ML skills for 100+ students by co-founding GDG On Campus at COMSATS Wah and organizing 5+ workshops and seminars on AI/ML, data science, and software development
- Strengthened project teams' skills in Python, model training, and evaluation metrics (precision, recall, F1-score) by mentoring across supervised learning and computer vision tracks
- Strengthened practical AI/ML skills for junior cohorts by delivering sessions on data preprocessing, feature engineering, and CNNs

DevOps & AI Intern | SPS - Software Productivity Strategists

Jul 2025 – Sep 2025

- Streamlined build-to-deploy workflows across development and staging environments by building CI/CD pipelines in Jenkins with automated triggers, test execution, and Slack/email notifications
- Improved service scalability and uptime by containerizing services with Docker and Docker Swarm, maintaining health via container monitoring and restart policies

EDUCATION

Bachelor of Science in Artificial Intelligence (BSAI)

Expected Sep 2027

COMSATS University Islamabad, Wah Campus | CGPA: **3.85/4.00**

Relevant Coursework: Artificial Neural Networks & Deep Learning, Machine Learning, Data Structures, Database Systems, Knowledge Representation & Reasoning, Design & Analysis of Algorithms, Linear Algebra

PROJECTS

Sortd | React, Vite, Node.js, Express, Gemini 1.5 Flash, Groq Whisper v3, Supabase (PostgreSQL), PWA

- Won 1st Place at HACKDATA V1 (Team MadGroot) by building a full-stack PWA that captures and organizes Instagram/YouTube content, using Gemini 1.5 Flash for multilingual (English, Urdu, Arabic) OCR, classification, and summarization on a React/Node.js + Supabase (PostgreSQL) stack
- Built an automated transcription pipeline (yt-dlp, ffmpeg, Groq Whisper v3) with real-time status via Server-Sent Events, and shipped one-tap Android/iOS content capture using the PWA Share Target API

Facial Expression Recognition Research Collaboration | University of Macau, PyTorch, I-Center Method, FER+, RAF-DB

- Improved emotion-recognition performance by +3.89pp macro-F1 over the state-of-the-art baseline (10-seed test, 95% CI positive) by validating the I-Center method across FER+ and RAF-DB datasets — with additional exploratory testing on ExpW — for a University of Macau research collaboration

BSL Hand Gesture Recognition System | TensorFlow, MobileNetV2, Transfer Learning, Streamlit

- Classified British Sign Language gestures at 95.74% test accuracy and 95.78% F1-score by fine-tuning MobileNetV2 with layer-wise transfer learning across 34,000 labeled images (34 classes)

- Delivered live webcam gesture recognition by building a real-time OpenCV inference pipeline deployed as an interactive Streamlit web app
- ThreatLens** | *ResNet50, TensorFlow, Keras, Transfer Learning, Django REST, React*
- Classified malware into 25 families by converting binary executables to grayscale images (byte-to-image technique) and training ResNet50 via transfer learning on the Maling dataset
 - Enabled end-to-end malware prediction with confidence scoring by building a Django REST API for file upload and inference, connected to a React frontend

- Retail Sales Performance Analytics Dashboard** | *Python, Pandas, SQL, ETL, EDA, Tableau*
- Processed 500,000+ UK retail transactions into revenue KPIs, customer segments, and seasonal trends by building a Python + Pandas ETL pipeline with SQL aggregations, GROUP BY queries, and window functions
 - Surfaced top product categories and high-value customer segments for BI reporting by designing interactive Tableau dashboards with calculated fields and parameter controls

CERTIFICATIONS

- AI Engineer Track — Core (LLM Engineering, RAG, QLoRA, Agents), Production, and Agentic (MCP) · Ligency, Ed Donner *In Progress*
- Machine Learning Specialization (Supervised ML & Advanced Learning Algorithms) · DeepLearning.AI — Andrew Ng *Sep 2025*
- Data Analysis with Python · IBM *Sep 2025*
- Python for Data Analysis: Pandas & NumPy · Coursera *Jul 2025*
- Git with GitLab & Bitbucket · LearnKartS *Jul 2025*
- Introduction to Programming with Python (CS50P) · Harvard University *Aug 2024*

TECHNICAL SKILLS

- Languages:** Python, SQL, C++, Java, PHP
- AI / ML & Deep Learning:** Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation (Precision, Recall, F1, ROC-AUC), CNNs, RNNs/LSTMs, Transfer Learning, MobileNetV2, ResNet50, TensorFlow, Keras, PyTorch
- LLM & RAG Systems:** RAG Pipelines, Hybrid Retrieval, Cross-Encoder Reranking, RAGAS Evaluation, Agentic AI, Vector Databases, Prompt Engineering
- Data & Analytics:** NumPy, Pandas, Scikit-learn, EDA, ETL Pipelines, SQL Aggregations & Window Functions, Tableau, KPI Dashboards
- MLOps & Deployment:** Git/GitHub, Docker, CI/CD (Jenkins), REST APIs (Django, Flask), Streamlit, PostgreSQL/Supabase, Vercel/Railway

ACHIEVEMENTS & ACTIVITIES

- 1st Place - HACKDATA V1 Hackathon (Team MadGroot) - Sortd AI Content Capture Platform
- 1st Place - Pitch Perfect & AI Model Training Competitions, HITEC Mega Code War 2026
- Academic Excellence Award - COMSATS University Wah Campus
- 1st Place - CS Quiz Competition, UET Taxila
- 1st Place - Speed Programming Competition, HITEC University
- Ranked 4th out of 80+ participants - Tech Quiz Competition, Tech Fest COMSATS Islamabad

LANGUAGES

- English (Professional) · Urdu (Native) · Pashto (Conversational)